

Charts

What is a Chart?

A **chart** is a graphical representation of data.

Instead of looking at numbers in a table:

Month Sales

Jan 10000

Feb 12000

Mar 9000

Apr 15000

A chart displays the data visually, making it much easier to identify increases, decreases, and patterns.

Why are Charts Important?

Data analysts use charts to:

- Present reports
- Identify trends
- Compare values
- Show percentages
- Detect patterns
- Make business decisions

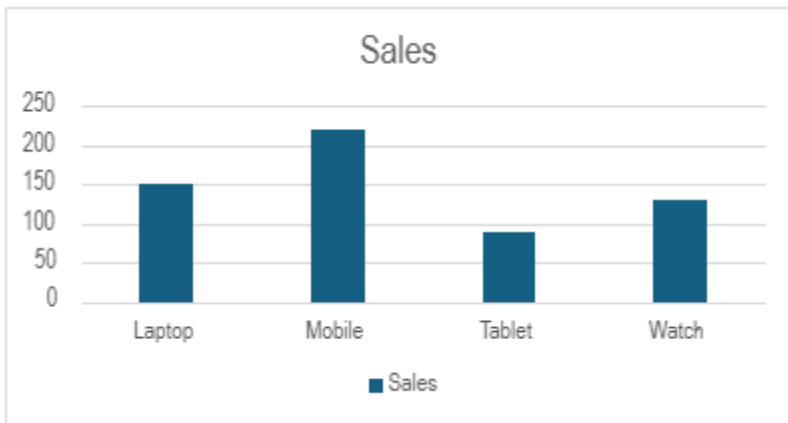
For example:

- Which month had the highest sales?
 - Which product sells the most?
 - Which region has the lowest revenue?
-

Types of Charts in Excel

There are many chart types, but these are the ones every data analyst should know.

1. Column Chart



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Purpose

Compare different categories.

Example:

Product Sales

Laptop 150

Mobile 220

Tablet 90

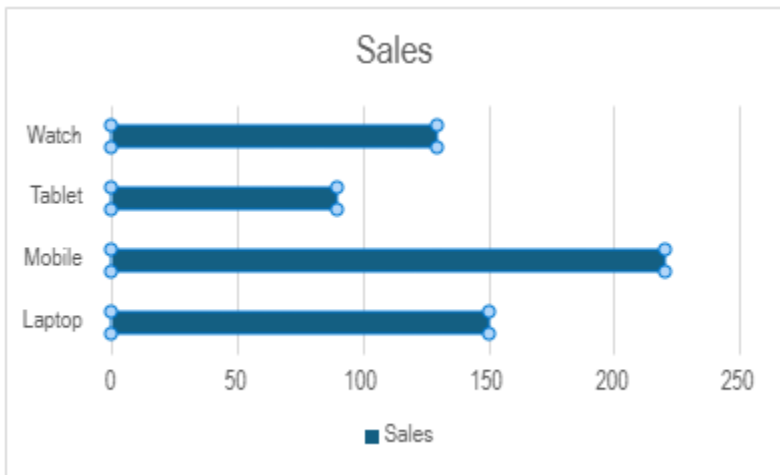
Watch 130

Each product is shown as a vertical bar.

Best for:

- Product comparison
- Employee performance
- Sales comparison
- Department comparison

2. Bar Chart



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Looks similar to a column chart, but the bars are horizontal.

Best for:

- Long category names
- Ranking
- Comparing many items

Example:

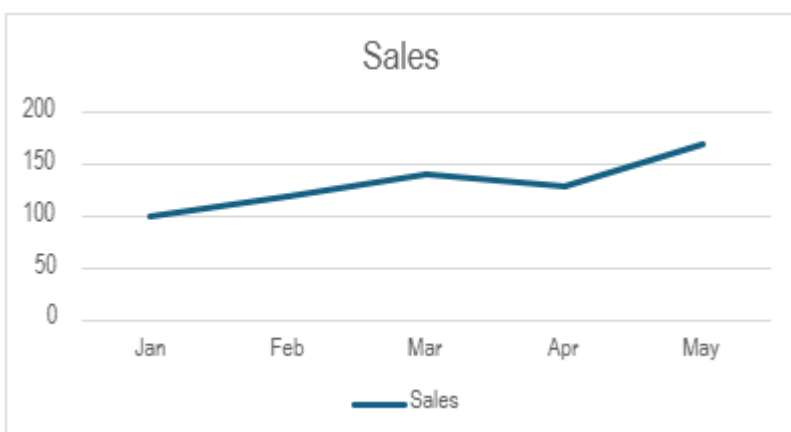
Employee Performance

John ————— 95

Anita ————— 120

Rahul ————— 80

3. Line Chart



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Purpose

Shows trends over time.

Example:

Month Sales

Jan 100

Feb 120

Mar 140

Apr 130

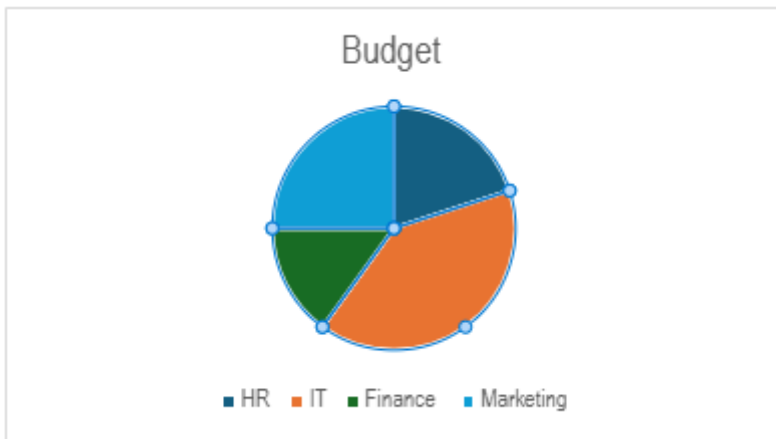
May 170

A line connects each month's sales.

Best for:

- Monthly sales
- Stock prices
- Temperature
- Website visitors

4. Pie Chart



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Purpose

Shows parts of a whole (percentages).

Example:

Department Budget

HR 20%

IT 40%

Finance 15%

Marketing 25%

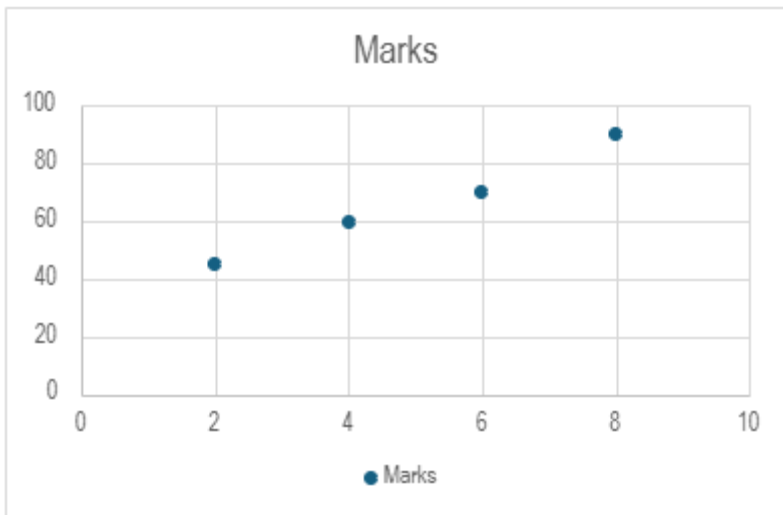
Each slice represents a percentage of the total.

Best for:

- Market share
- Budget allocation
- Survey responses

Avoid using a pie chart when there are many categories.

5. Scatter Chart



6

Purpose

Shows the relationship between two numbers.

Example:

Study Hours Marks

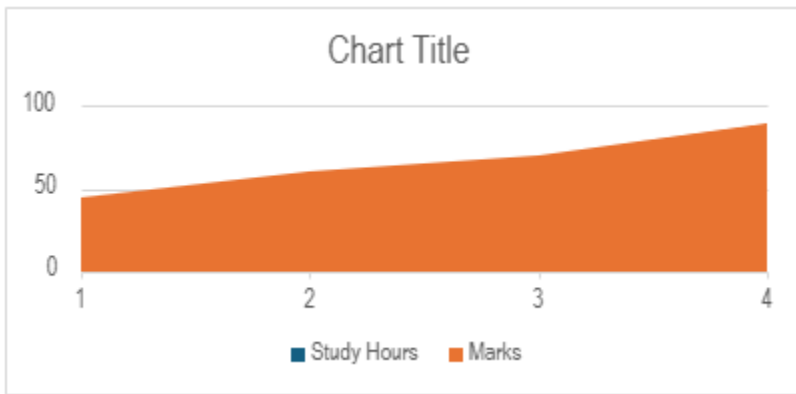
2	45
4	60
6	70
8	90

The chart helps identify whether studying more is associated with higher marks.

Best for:

- Correlation
- Regression
- Statistical analysis

6. Area Chart



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Similar to a line chart, but the area beneath the line is filled.

Best for:

- Showing totals over time
- Displaying cumulative values